

Pure-dyadic regulator-uniform neighbourhood gap boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope: a cylinder-free uniform local radius

R-156 proved a nonzero shifted-state neighbourhood only after fixing one finite cutoff. R-159 removes that auxiliary-cylinder dependence for the single centered pure-dyadic chart. It does so without choosing a preferred cutoff or multiplier and without turning a qualitative radius into a numerical one.

Fix the side-16 torus, the hash-pinned A1 symbol, the A7 floor $\varepsilon_\rho = 10^{-12}$, and exact continuum torus integration. For each cutoff use the same real-even scalar multiplier on every real component, with

$$|m_\Lambda(k)| \leq 1, \quad m_\Lambda(-k) = m_\Lambda(k), \quad (1.1)$$

and subtract that regulator's exact primitive derivative covariance. Let $p \neq 0$ be a dual-lattice momentum for which $p, 2p, 4p$ are retained. The raw both-sign roots $\xi_p, \xi_{2p}, \xi_{4p}$ are centered, whitened and mutually independent. There is no deterministic past mean. Define

$$\chi(A, B) = (A\xi_p, B(\xi_{2p} + A\xi_p)), \quad A, B \in \mathbb{R}^{12 \times 12}. \quad (1.2)$$

Theorem 1.1 (regulator-uniform local gap). There exists $\delta_* > 0$, depending only on the fixed torus, A1 symbol, positive floor and multiplier bound in (1.1), but independent of the cutoff, the particular admitted regulator and the retained p , such that

$$\|A\|_{\text{HS}} + \|B\|_{\text{HS}} < \delta_* \quad (1.3)$$

implies, for every nonzero coefficient direction $v = (H, G)$,

$$D^2(\mathcal{A} \circ \chi)_{A,B}[v, v] > \frac{1}{10}(\|H\|_{\text{HS}}^2 + \|G\|_{\text{HS}}^2). \quad (1.4)$$

The intrinsic physical-source Hessian on the tangent image also obeys

$$D^2\mathcal{A}(h_{A,B})[D\chi v, D\chi v] > \frac{1}{10}\mathcal{G}_{A,B}(v, v), \quad (1.5)$$

where

$$\mathcal{G}_{A,B}(v, v) = \|H\|_{\text{HS}}^2 + \|G\|_{\text{HS}}^2 + \|GA + BH\|_{\text{HS}}^2. \quad (1.6)$$

The radius is existential. No numerical value, floor-uniform statement, finite-grid alias theorem, realised-past fibrewise result, growing multiroot union, arbitrary predictable nonlinear or revisit estimate, T-050 closure, or Sector-A closure is claimed.

2. The nonlinear controller still has a centered Gaussian endpoint

Write $S_r(x)$ for the regulator-weighted whitened synthesis at the antipodal pair $\{rp, -rp\}$ and $D_{r,i} = \partial_i S_r$, for $r = 1, 2, 4$. The endpoint columns after (1.2) are

$$\begin{aligned} R_1 &= S_1 + S_2 A + S_4 B A, & R_2 &= S_2 + S_4 B, & R_4 &= S_4, \\ U_{1,i} &= D_{1,i} + D_{2,i} A + D_{4,i} B A, & U_{2,i} &= D_{2,i} + D_{4,i} B, & U_{4,i} &= D_{4,i}. \end{aligned} \quad (2.1)$$

Thus the chart is nonlinear in (A, B) but the endpoint is linear in the raw Gaussian vector. In particular the field W and currents V_i remain jointly centered Gaussian for every deterministic pair (A, B) .

Let C_{bg} and Q_{bg} be the background value and derivative covariances. Then

$$\begin{aligned} C &= C_{\text{bg}} + \sum_{r=1,2,4} R_r R_r^*, \\ Q &= Q_{\text{bg}} + \sum_i \sum_{r=1,2,4} U_{r,i} U_{r,i}^*, \\ K_i &= \sum_{r=1,2,4} R_r U_{r,i}^*. \end{aligned} \quad (2.2)$$

Common evenness gives $K_i(0, 0) = 0$. Put

$$Q_0 = Q(0, 0). \quad (2.3)$$

This Q_0 is exactly the primitive covariance used by the A7 subtraction.

2.1 All-state covariance-normal Gaussian identity

Let $\mathcal{B}(w) = C_6(w)^* C_6(w)$ be the fixed six-row coefficient matrix. For any centered jointly Gaussian (W, V_i) , Gaussian integration by parts gives

$$\mathbb{E}[(V_i)_m (V_i)_n \mathcal{B}_{mn}(W)] = Q_{i,mn} \mathbb{E} \mathcal{B}_{mn}(W) + (K_i)_{am} (K_i)_{bn} \mathbb{E} \partial_{ab} \mathcal{B}_{mn}(W). \quad (2.4)$$

Consequently the complete covariance-normal current endpoint satisfies, for every (A, B) ,

$$2\mathcal{R}(A, B) = \int \left\{ \text{Tr}[(Q - Q_0) \bar{\mathcal{B}}(C)] + \sum_i (K_i)_{am} (K_i)_{bn} \bar{T}_{abmn}(C) \right\} dx, \quad (2.5)$$

where

$$\bar{\mathcal{B}}(C) = \mathbb{E}_{N(0, C)} \mathcal{B}(W), \quad \bar{T}(C) = \mathbb{E}_{N(0, C)} D^2 \mathcal{B}(W). \quad (2.6)$$

Equation (2.5), rather than the origin-only R-151 Hessian formula, is the new load-bearing lemma. Differentiating the complete scalar in (2.5) includes all first and second C, Q, K jets and exactly the projected-force pairing with the chart acceleration. Source, trace, current, forward, legal reverse and balanced descriptions are not added again.

3. Exact cancellation before compactness

The raw derivative-covariance family is not uniformly bounded. For the positive-density sharp-cube member, the A1 fourth-order symbol gives

$$\text{Tr } Q_0^{\text{sharp}} \asymp \sum_{|k| \leq \Lambda} \frac{|k|^2}{1 + |k|^4} = \Theta(\Lambda). \quad (3.1)$$

Other admitted multipliers can have smaller or scheme-dependent growth; the class as a whole is nevertheless not precompact in raw Q . Thus any proof which first puts the whole raw- Q family in a compact parameter set is false. Equation (2.5) instead cancels Q_0 algebraically. The difference $\Delta Q = Q - Q_0$ contains only the finitely many controlled $p, 2p, 4p$ columns.

R-151's exact lower symbol polynomial is

$$f(x) = x^2 - \frac{4626377063}{5000000000}x + \frac{5020336473}{10000000000}. \quad (3.2)$$

Its continuous minimum is

$$\nu = \frac{28800000000947494031}{10^{20}} > 0. \quad (3.3)$$

For $r \in \{1, 2, 4\}$ and $V = 16^3$,

$$\|S_r\|_{\text{op}}^2 \leq \frac{2}{V f(r^2|p|^2)} \leq \frac{2}{V\nu}, \quad (3.4)$$

and

$$\|D_r\|_{\text{op}}^2 \leq \frac{2r^2|p|^2}{V f(r^2|p|^2)} \leq \frac{2}{V} \sup_{y \geq 0} \frac{y}{f(y)} < \infty. \quad (3.5)$$

The multiplier bound in (1.1) can only decrease these norms. The value and derivative syntheses vanish respectively as $|p|^{-2}$ and $|p|^{-1}$ at large momentum. On any fixed controller ball, (2.1) therefore gives uniform polynomial bounds for C , ΔQ , K_i and all controller jets needed by a Hessian.

The background value covariance is harmless:

$$0 \preceq C_{\text{bg}} \preceq C_{\text{val}} \text{Id}, \quad C_{\text{val}} \lesssim \frac{1}{V} \sum_k \langle k \rangle^{-4} < \infty. \quad (3.6)$$

Its possible values have compact closure in the finite-dimensional positive semidefinite cone.

4. Finite invariants and the uniform modulus

The controller degrees are finite and fixed:

object	χ	R, U	$C, \Delta Q, K$	$K \otimes K$	$\mathcal{C}_{\text{CM}}$	$\mathbb{E} W ^6$
maximum degree	2	2	4	8	4	12.

(4.1)

The Cameron–Martin term is exactly

$$\mathcal{C}_{\text{CM}} = \frac{9}{20} \left(\|A\|_{\text{HS}}^2 + \|B\|_{\text{HS}}^2 + \|BA\|_{\text{HS}}^2 \right), \quad (4.2)$$

and for a centered six-real Gaussian

$$\mathbb{E}|W|^6 = (\text{Tr } C)^3 + 6 \text{Tr } C \text{Tr}(C^2) + 8 \text{Tr}(C^3). \quad (4.3)$$

At the fixed positive floor every derivative of the rational matrix $\mathcal{B}(W)$ has polynomial growth. The maps

$$C \longmapsto \mathbb{E}_{N(0,C)} D^j \mathcal{B}(W) \quad (4.4)$$

are therefore continuous on bounded positive semidefinite covariance sets, including singular covariances. This derivative-expectation representation is essential: the alternative score formula containing C^{-1} is not uniform when a regulator makes a covariance eigenvalue tend to zero.

The remaining parameters have compact closure after the following exact reductions: C_{bg} lies in (3.6); the three controlled multiplier values lie in $[-1, 1]$; the momentum direction lies in S^2 ; and the radial variable $|p|^2$ is compactified by adjoining its infinite limit, where every controlled synthesis vanishes. All

carriers are the collinear multiples $p, 2p, 4p$. For every nonzero dual-lattice p , Haar measure under $x \mapsto p \cdot x$ is uniform on the circle, so exact spatial integration is one common compact phase average. We do not assert that the oscillatory functions themselves form a compact family in a strong function norm.

Let $J_{A,B} = D\chi_{A,B}$. R-156 gives

$$\mathcal{G} = J^* J = \text{Id} + T^* T \succeq \text{Id}, \quad (4.5)$$

where $T(H, G) = GA + BH$. On a bounded controller ball, both $\mathcal{G}$ and $\mathcal{G}^{-1/2}$ are uniformly continuous. No physical covariance inverse is used.

Write $\mathcal{H}_{\text{ctl}}$ for the complete pullback Hessian and

$$\hat{\mathcal{H}}_{\text{int}} = \mathcal{G}^{-1/2} J^* H_\pi(h) J \mathcal{G}^{-1/2} \quad (4.6)$$

for the normalized intrinsic Hessian. Define

$$\omega(r) = \sup \max \{ \|\mathcal{H}_{\text{ctl}}(A, B) - \mathcal{H}_{\text{ctl}}(0, 0)\|_{\text{op}}, \|\hat{\mathcal{H}}_{\text{int}}(A, B) - \hat{\mathcal{H}}_{\text{int}}(0, 0)\|_{\text{op}} \}, \quad (4.7)$$

where the supremum is over (1.1), all admitted cutoffs and retained p , and $\|A\|_{\text{HS}} + \|B\|_{\text{HS}} \leq r$. The repaired compact finite-invariant reduction proves

$$\omega(r) \longrightarrow 0 \quad (r \downarrow 0). \quad (4.8)$$

5. Uniform gap and owner-complete two-block meaning

R-155 supplies, uniformly over the pure-dyadic origin family,

$$\mathcal{H}_{\text{ctl}}(0, 0) = \hat{\mathcal{H}}_{\text{int}}(0, 0) \succeq \frac{147}{1000} \text{Id}. \quad (5.1)$$

The exact headroom over the T-050 comparison target is

$$\frac{147}{1000} - \frac{1}{10} = \frac{47}{1000}. \quad (5.2)$$

Choose $\delta_* > 0$ so that

$$\omega(\delta_*) < \frac{47}{2000}. \quad (5.3)$$

Then both complete Hessians retain the comparison value

$$\frac{147}{1000} - \frac{47}{2000} = \frac{247}{2000} > \frac{1}{10}, \quad (5.4)$$

which proves Theorem 1.1.

In the actual two-source decomposition, subtracting the target metric gives one self-adjoint block operator

$$N = M - \frac{1}{10} \mathcal{G} = \begin{pmatrix} E & C \\ C^* & F \end{pmatrix}. \quad (5.5)$$

The forward entry is C , its sole legal reverse is C^* , and the balanced cross is $2 \text{Re}\langle x, Cy \rangle$. These are three descriptions of one Hessian block, not three payments. The source, covariance-normal current, trace and sextic terms occur once in the scalar differentiated above. The projected-force/controller connection also occurs once through the second derivative of (2.5). No independent low coordinate exists in (1.2); an invented low/Feshbach reserve remains prohibited. Thus R-159 supplies the requested unified forward/legal-reverse/balanced/no-independent-low estimate on this local chart, but not on the full arbitrary predictable source class.

6. Failure modes and scope firewall

1. **Raw- Q compactness is false.** Equation (3.1) diverges. The proof is valid only after the exact $Q - Q_0$ cancellation in (2.5).
2. **A covariance-inverse score bound is not uniform.** Admitted multiplier sequences can make a value covariance singular. The proof uses $\mathbb{E}_C D^j B$, never C^{-1} .
3. **Bounded oscillatory functions are not a compact family.** The theorem uses the exact Haar phase reduction for the collinear $p, 2p, 4p$ carriers, not a false strong-function compactness argument.
4. **Generic finite-grid de-aliasing is not covered.** A grid on which, for example, $3p$ aliases to zero can destroy the origin Fourier cancellation. The theorem assumes exact continuum torus integration; a numerical grid needs a separate certified alias error.
5. **The entire unnormalised A7 multiplier union is not covered.** The single radius is proved for the contraction subclass $|m| \leq 1$. A class with a separately fixed envelope M would give a radius depending on M .
6. **The radius is not effective.** An explicit value requires registered bounds for higher coefficient derivatives, Gaussian moments, all covariance jets, the sextic third variation, projected force and graph normalization. R-130 and A7 prove finiteness but do not pin all those constants.
7. **The chart is not the full predictable class.** A nonzero deterministic mean, arbitrary nonlinear function of a realised past, a growing simultaneous root union or a general revisit graph is outside the compact parameter family.
8. **No phase or equation is selected.** No BCC ansatz, vacuum comparison, candidate PDE or morphology discriminator enters this theorem.

7. Devil's-advocate review

1. **Objection: R-151 only proved an origin Hessian. VALID WITH REPAIR.** R-159 proves the all-state identity (2.5) before differentiation. It does not extend the origin formula by assertion.
2. **Objection: the projected-force connection was omitted. DISMISSED.** It is exactly the chain-rule part of the second controller derivative of the one scalar (2.5).
3. **Objection: the derivative covariance destroys uniformity. VALID FOR RAW Q , DISMISSED AFTER NORMAL ORDERING.** Q_0 diverges, but cancels identically. Every controlled ΔQ jet is uniformly bounded by (3.4)–(3.5).
4. **Objection: singular value covariance invalidates the proof. DISMISSED.** Fixed-floor coefficient derivatives have polynomial growth and their Gaussian expectations extend continuously to singular positive semidefinite covariance. No score inverse is used.
5. **Objection: compactness silently chooses a production cylinder. DISMISSED.** The theorem is universally quantified over the declared contraction regulator class. Compactness is applied to the closure of finite invariants after exact cancellation and phase reduction.
6. **Objection: a uniform local radius closes T-050. UPHELD AGAINST THAT INFERENCE.** General predictable functions, realised past, growing source unions and arbitrary revisit structures remain outside R-159.

External review is invited for the all-state Gaussian factor in (2.4), the orientation of K_i , exact cancellation of every Q_0 copy, root realification factors, Haar phase pushforward, singular-covariance continuity, the source-Gram normalization, the half-headroom comparison, and the exact-integration scope firewall.

8. Reproduction, next boundary, and Result footer

From the repository root, run

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_pure_dyadic_regulator_uniform_neighborhood_gap_boundary.py
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_pure_dyadic_regulator_uniform_neighborhood_gap_boundary_independent.py
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_pure_dyadic_regulator_uniform_neighborhood_gap_boundary_verify.py
```

R-159 corrects the operational reading of EXP-000685: absence of a uniquely pinned finite cylinder blocks a unique numerical matrix, but it does not block a theorem quantified over the bounded regulator class. The next T-050 boundary is no longer this local cylinder. It is the passage from one centered finite Gaussian chart to growing multiroot unions and arbitrary predictable nonlinear/revisit controls. An explicit radius package may be useful as a robustness audit, but it is not the logical closure gate.

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Result ID: A13-CLASSII-PURE-DYADIC-REGULATOR-UNIFORM-
NEIGHBORHOOD-GAP-BOUNDARY (R-159).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: One existential delta_* works for every admitted cutoff,
|m|<=1 common-even regulator and retained p:2p:4p centered shifted chart,
and preserves both pullback and intrinsic normalized gaps above 1/10.
Scope: Fixed L=16/A1 symbol/floor; exact continuum torus integral; centered
single pure-dyadic chart; no numerical radius or general predictable class.
Dependencies: A1, A7, R-150, R-151, R-155, and R-156 manifests hash-pinned.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_pure_dyadic_regulator_uniform_
neighborhood_gap_boundary_verify.py
Expected output: Primary, non-importing independent and integrated PASS.
Falsification gate: Any all-state Gaussian identity, Q0 cancellation, root
bound, compact phase reduction, source normalization, gap arithmetic,
authority, artifact, PDF, public-surface or release check fails.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: No floor removal, alias theorem, realised-past,
growing multiroot, arbitrary nonlinear/revisit, T-050/A13, Nelson/measure,
phase/PDE, or Sector-A closure follows.
Next required action: Test the growing-union/arbitrary-predictable boundary
or produce an intrinsic counterexample; keep truth-first T-054 primary.
```